

A — Gradient Descent Regression

Maths for Machine Learning — Group Mini-Project

International MSc Finance & Data — Albert School

2026

Outline

Problem & Dataset

Exploratory Data Analysis

Mathematical Framework

Learning Rate & Convergence

Visualising Gradient Descent

GD Variants & Comparison

Ridge Regularisation (Session 5)

PCA (Session 1)

Conclusions

Problem & Dataset

The task

Goal: predict house sale prices using linear regression.

- Implement gradient descent **from scratch** (no sklearn).
- Tune learning rates, compare convergence.
- Benchmark against closed-form OLS.
- Explore with PCA and Ridge regression.

Core sessions: all five (S1–S5).

Dataset

Kaggle — Ames Housing
1 460 houses $\times$ 80 columns

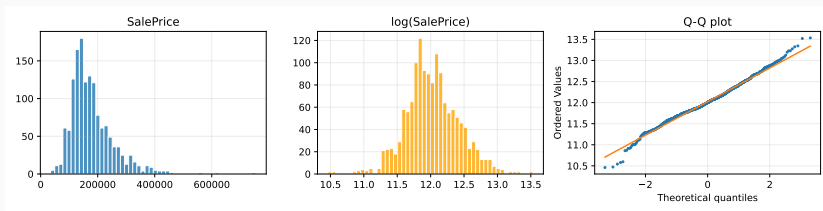
Selected **10 numeric features**:

OverallQual, GrLivArea,
GarageCars, GarageArea,
TotalBsmtSF, 1stFlrSF,
FullBath, TotRmsAbvGrd,
YearBuilt, YearRemodAdd

Target: $\log(\text{SalePrice})$

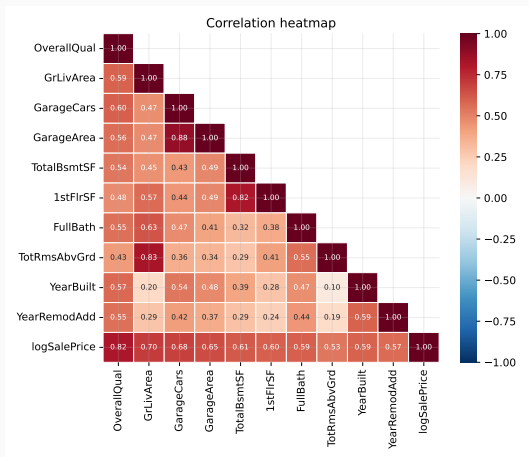
Exploratory Data Analysis

Target distribution & normality (Session 3)



Raw skew = 1.88 $\rightarrow$ log skew = 0.12. Q-Q plot confirms near-normality.
Shapiro-Wilk $p < 0.05$ but visual improvement is clear — justifies log-transform for MSE.

Correlation heatmap



Session 1: correlation = $\frac{\vec{x}_i \cdot \vec{x}_j}{\|\vec{x}_i\| \|\vec{x}_j\|}$ —
normalised **dot product**.

Strongest predictors:

Feature	r
OverallQual	+0.82
GrLivArea	+0.71
GarageCars	+0.65
TotalBsmtSF	+0.62

Note: GarageCars/GarageArea are highly correlated ($r = 0.88$) — PCA will handle this redundancy.

Hypothesis test — does quality matter? (Session 4)

Welch's t -test (unequal variances):

$$H_0: \mu_{\text{high}} = \mu_{\text{low}}$$

$$H_1: \mu_{\text{high}} \neq \mu_{\text{low}}$$

$$t = 28.2, \quad p \approx 10^{-116}$$

Reject H_0 at $\alpha = 0.05$.

OverallQual is a statistically significant predictor — confirmed by both correlation and hypothesis testing.



Mathematical Framework

The model — Sessions 1 & 2

Data as matrix: $X \in \mathbb{R}^{n \times d}$, each row = one house.

Closed-form OLS (Session 1 — matrix inverse):

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

MSE loss (Session 2 — gradient descent):

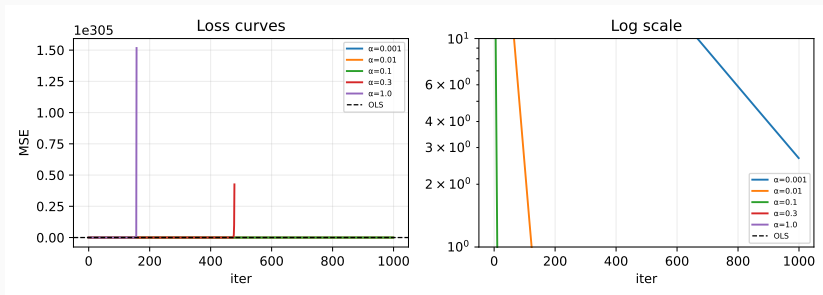
$$\mathcal{L}(\vec{w}) = \frac{1}{n} \|y - X\vec{w}\|^2, \quad \nabla_{\vec{w}} \mathcal{L} = \frac{-2}{n} X^T (y - X\vec{w})$$

Update rule: $\vec{w}_{t+1} = \vec{w}_t - \alpha \nabla \mathcal{L}(\vec{w}_t)$

Results: both methods $\rightarrow$ Test $R^2 = 0.80$, $\text{MSE} = 0.032$

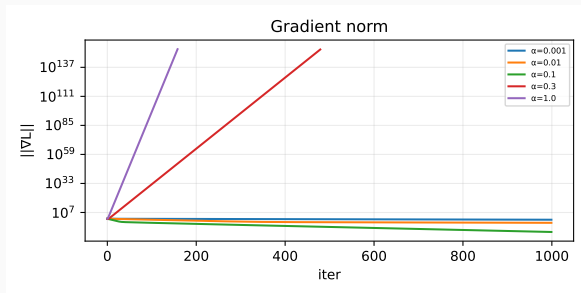
Learning Rate & Convergence

Learning rate sweep



$\alpha = 0.001$: too slow. $\alpha = 0.01, 0.1$: converge. $\alpha = 0.3, 1.0$: **diverge**.

Gradient norm



$\|\nabla \mathcal{L}\| \rightarrow 0$ at convergence. For $\alpha \geq 0.3$, it explodes.

Why? The Hessian eigenvalue bound

Session 1 × **Session 2**. The Hessian is: $H = \frac{2}{n} X^\top X$

Its **eigenvalues** (Session 1) set the curvature. Stability requires:

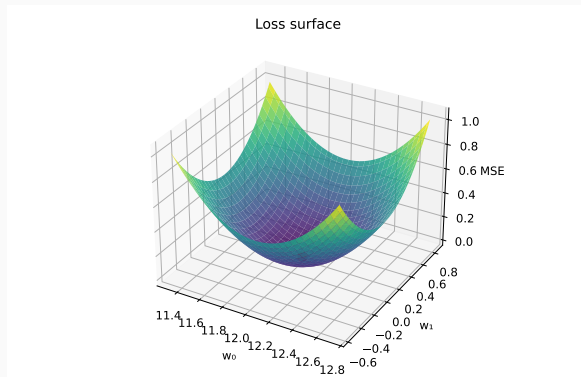
$$\alpha < \frac{2}{\lambda_{\max}(H)} = \frac{2}{10.27} = 0.195$$

	Value	Verdict
$\alpha = 0.1$	< 0.195	safe
$\alpha = 0.3$	> 0.195	diverges
$\alpha = 1.0$	$\gg 0.195$	diverges

Condition number $\kappa = 50$ — standardisation reduced this by orders of magnitude.

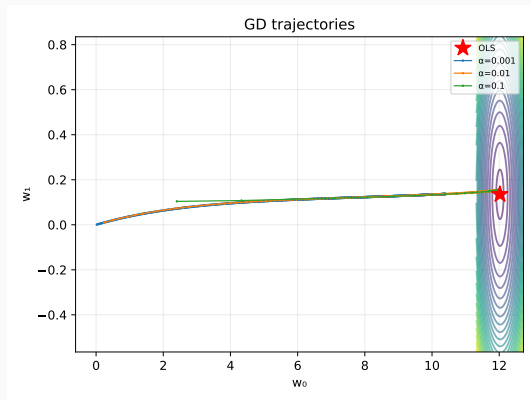
Visualising Gradient Descent

3D loss surface



Slice through w_0 (bias) and w_1 (OverallQual). MSE is a **convex bowl**.

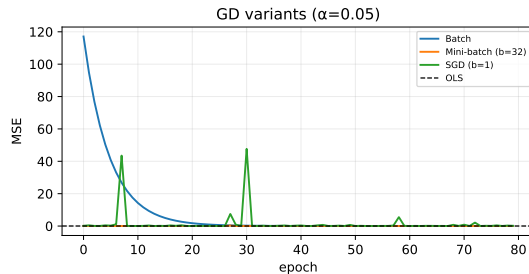
GD trajectories on contour



Paths from $\vec{w} = \vec{0}$ toward the OLS minimum. Larger α = bigger steps.

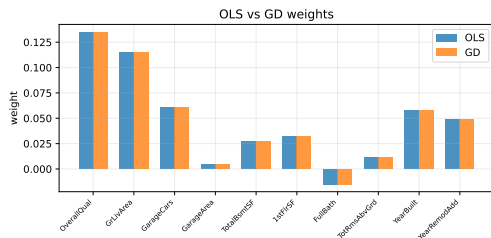
GD Variants & Comparison

Batch vs Mini-batch vs SGD



Variant	Samples/step	Trade-off
Batch	all n	Exact, slow
Mini-batch	$b = 32$	Balanced
SGD	1	Noisy, fast

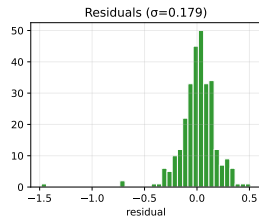
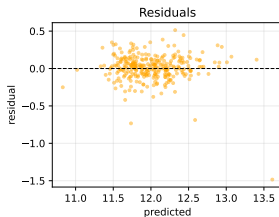
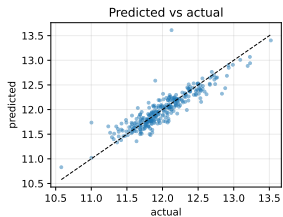
GD vs OLS — head-to-head



Metric	OLS	GD
Test MSE	0.032	0.032
Test R^2	0.802	0.802
$\ \vec{w}_{GD} - \vec{w}_{OLS}\ $	—	10^{-15}

GD *exactly* recovers OLS to machine precision with a safe α .

Test predictions & residuals

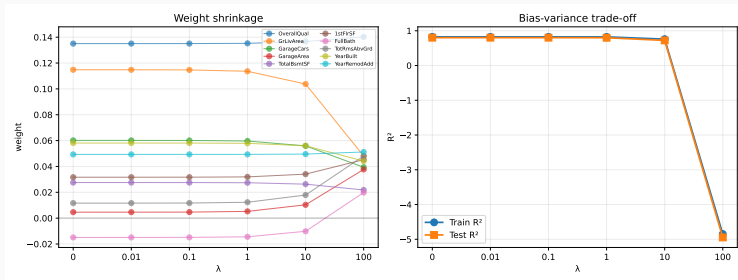


Ridge Regularisation (Session 5)

Ridge — L_2 penalty

Add a regularisation term:

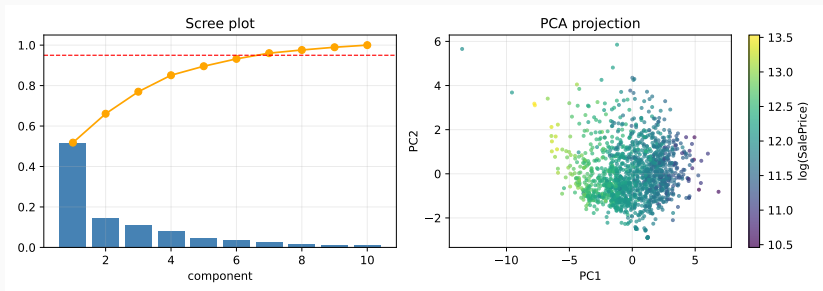
$$\mathcal{L}_{\text{Ridge}} = \frac{1}{n} \|y - X\vec{w}\|^2 + \frac{\lambda}{n} \|\vec{w}\|^2$$



Moderate λ (≤ 1) maintains $R^2 \approx 0.80$. Large λ over-regularises.

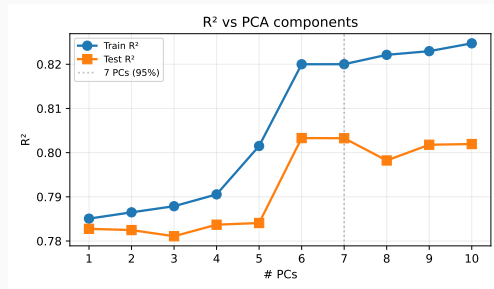
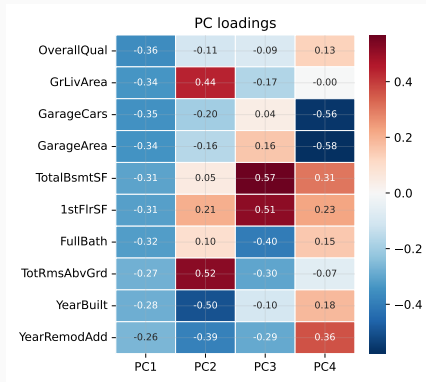
PCA (Session 1)

PCA on house features



PC1 (52% variance) = “house quality” axis. Colour confirms: PC1 sorts by price.

PC loadings & R^2 vs components



7 of 10 PCs capture 95% variance.

Top-7 PCs: $R^2 = 0.803$ vs all-features 0.802.

30% dimensionality reduction with $< 0.2\%$ R^2 loss.

Conclusions

Key takeaways

1. **GD matches OLS** to 10^{-15} precision with the right α .
2. **Hessian eigenvalue bound** $\alpha < 2/\lambda_{\max}$ predicts divergence — eigenvalues (S1) govern optimisation (S2).
3. **Standardisation** compressed κ by orders of magnitude.
4. **PCA** achieves 30% reduction with minimal R^2 loss.
5. **Ridge** offers a natural GD extension for complexity control.

Session	Concept	Application
1 — Linear Algebra	Eigenvalues, PCA, $X^T X$	OLS, Hessian, PCA
2 — Calculus	Gradient, chain rule	GD update, LR bound
3 — Probability	Distributions	Log-transform, Q-Q
4 — Statistics	Hypothesis testing	Welch's t -test
5 — Advanced	Regularisation	Ridge GD

Questions?